

Recursive Trace Geometry and Observer-Relative Spectral Structure

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Abstract

We investigate the relationship between recursive observer traces and emergent spectral geometry in hierarchical Markov systems. Building on observer-first approaches and recent discussions of recursive trace logic, we interpret observer windows as coarse-grained dynamical traces over hidden-state Markov processes. We then construct induced Laplacian geometries from these traces and study their spectral properties using diffusion-based spectral dimension measures.

The central result is that observer traces generate effective geometries distinct from ordinary visible-state subgraphs. In particular, hidden-state excursions of the form $B(I - C)^{-1}D$ alter diffusion structure and effective spectral dimension. A sequence of four toy models is presented: a baseline grid trace, a hierarchical trace system, a robustness/control experiment, and a direct trace-versus-subgraph comparison. The final comparison shows that hidden excursions are not equivalent to simple node deletion; they induce effective long-range connectivity in the observer's visible world.

These results do not derive spacetime, gravity, or quantum theory. They provide a toy-model framework in which observer-relative geometry can be studied as an emergent consequence of hidden dynamical organization.

1 Introduction

A growing number of approaches in theoretical physics suggest that spacetime may not be fundamental, but instead emerges from deeper relational or informational structures. Programs in quantum gravity, holography, tensor-network geometry, and observer-first approaches increasingly treat geometry as a derived rather than primitive concept [2, 5, 6].

Hoffman and Prakash's conscious-agent framework provides one formal observer-first approach in which observer dynamics are modeled using stochastic structures [1]. More recent publicly available seminar discussions by Hoffman and collaborators motivate a related idea: an observer window may be represented as a Markov trace of a larger process. In this picture, an observer does not access the full dynamics directly. The observer experiences an induced dynamics determined by visible states together with hidden excursions through inaccessible states.

At the same time, spectral geometry provides a natural language for describing emergent relational structure. Diffusion processes and Laplacian spectra are central tools in graph geometry, diffusion maps, causal dynamical triangulations, and network-based approaches to emergent spacetime [2, 7, 4, 3].

This paper connects these two directions. The guiding proposal is:

Recursive trace logic supplies the observer hierarchy; spectral geometry supplies the emergent spatial structure of each observer's world.

The central conceptual chain explored here is

$$P \longrightarrow \text{Tr}_A(P) \longrightarrow L_A \longrightarrow P_A(\sigma) \longrightarrow d_s^A(\sigma), \quad (1)$$

where P is a full Markov dynamics, $\text{Tr}_A(P)$ is the observer-window trace, L_A is the induced Laplacian, $P_A(\sigma)$ is the heat return probability, and $d_s^A(\sigma)$ is the observer-relative spectral dimension.

The aim is deliberately modest. We do not claim to derive spacetime, gravity, or quantum theory. We ask whether recursive observer traces naturally induce distinct spectral geometries, and whether hidden dynamical structure modifies those geometries in a way that can be measured.

2 Recursive Markov Traces

2.1 Markov Dynamics

We begin with a finite Markov chain represented by a row-stochastic transition matrix:

$$P_{ij} \geq 0, \quad \sum_j P_{ij} = 1. \quad (2)$$

Let the visible states accessible to an observer be denoted by A , while the remaining hidden states are grouped into a complementary sector. Reordering the transition matrix into visible and hidden blocks gives

$$P = \begin{pmatrix} A & B \\ D & C \end{pmatrix}. \quad (3)$$

Here A describes transitions among visible states, C describes hidden-state dynamics, B describes exits from visible to hidden states, and D describes re-entries from hidden to visible states.

2.2 The Trace Operation

The induced observer dynamics are given by the Markov trace formula

$$\text{Tr}_A(P) = A + B(I - C)^{-1}D. \quad (4)$$

The inverse term $(I - C)^{-1}$ encodes all possible hidden excursions through inaccessible states before re-entering the visible observer sector. This is the crucial distinction between recursive trace geometry and ordinary coarse-graining: the observer does not merely experience a restricted visible subgraph, but a visible dynamics modified by hidden excursions.

3 Induced Spectral Geometry

Given a trace observer dynamics

$$P_A = \text{Tr}_A(P), \quad (5)$$

we define an induced Laplacian geometry by

$$L_A = I - P_A. \quad (6)$$

In numerical simulations we use a symmetrized normalized form of the Laplacian to ensure a real spectrum suitable for diffusion geometry. For each observer dynamics we construct

$$L = I - D^{-1/2} P_A D^{-1/2}, \quad (7)$$

where W is the symmetrized transition matrix and D is the degree matrix.

The trace matrix $P_A = A + B(I - C)^{-1}D$ is generically asymmetric. The symmetrization step converts a directed random walk into an undirected diffusion geometry, losing directional flow information. This simplification is appropriate for the present exploratory analysis focused on spectral geometry, but future work should investigate directed Laplacian constructions preserving transition asymmetry.

The diffusion return probability is

$$P_A(\sigma) = \frac{1}{|A|} \text{Tr} \left(e^{-\sigma L_A} \right), \quad (8)$$

where σ is the diffusion scale and $|A|$ is the number of visible observer states. The observer-relative spectral dimension is then

$$d_s^A(\sigma) = -2 \frac{d \log P_A(\sigma)}{d \log \sigma}. \quad (9)$$

This is a running, scale-dependent dimension rather than a fixed dimension. The most informative regime is the intermediate diffusion-scale range, before small-scale discreteness and large-scale finite-size effects dominate.

4 Numerical Methods

4.1 Hierarchical Markov Construction

The numerical toy models consist of hierarchical Markov systems constructed from nested modular communities. The primary hierarchy contains 4 modules, 3 submodules per module, and 6 states per submodule, for a total of 72 states. Transition strengths are assigned hierarchically: strong coupling within submodules, weaker coupling within modules, weak coupling between modules, and small random perturbations to ensure ergodicity. The weighted adjacency structure is normalized row-wise to produce a stochastic Markov matrix.

Observer windows are defined at three scales: full-system observers, module-level observers, and submodule-level observers. For each observer window, two constructions are compared:

1. the recursive trace observer, computed using $\text{Tr}_A(P)$;
2. the ordinary visible-state subgraph observer, obtained by restricting to visible-visible transitions and renormalizing rows.

The comparison isolates the geometric effect of hidden-state dynamics.

Spectral dimensions were computed using diffusion scales

$$\sigma \in [10^{-1}, 10^{2.2}]. \quad (10)$$

4.2 Toy Model Sequence

Four progressively stronger toy models were used.

Toy 1. A baseline trace construction on a simple finite grid, demonstrating that a trace observer does not merely inherit a smaller copy of the full system.

Toy 2. A hierarchical Markov system comparing full-system, module-level, and submodule-level trace observers.

Toy 3. A multi-trial robustness/control experiment comparing hierarchical, modular-control, and random-control systems.

Toy 4. A direct comparison between recursive Markov traces and ordinary visible-state subgraphs, isolating the role of hidden excursions.

5 Toy Model Results

5.1 Toy 1: Baseline Trace Geometry

The first toy model used a simple finite Markov grid and compared the full dynamics to a smaller observer-window trace. The purpose was not to establish hierarchy, but to check the basic pipeline

$$P \rightarrow \text{Tr}_A(P) \rightarrow L_A \rightarrow d_s^A(\sigma).$$

The result showed that the trace observer does not merely see a smaller version of the full system. The hidden dynamics modifies the effective spectral profile of the observer window.

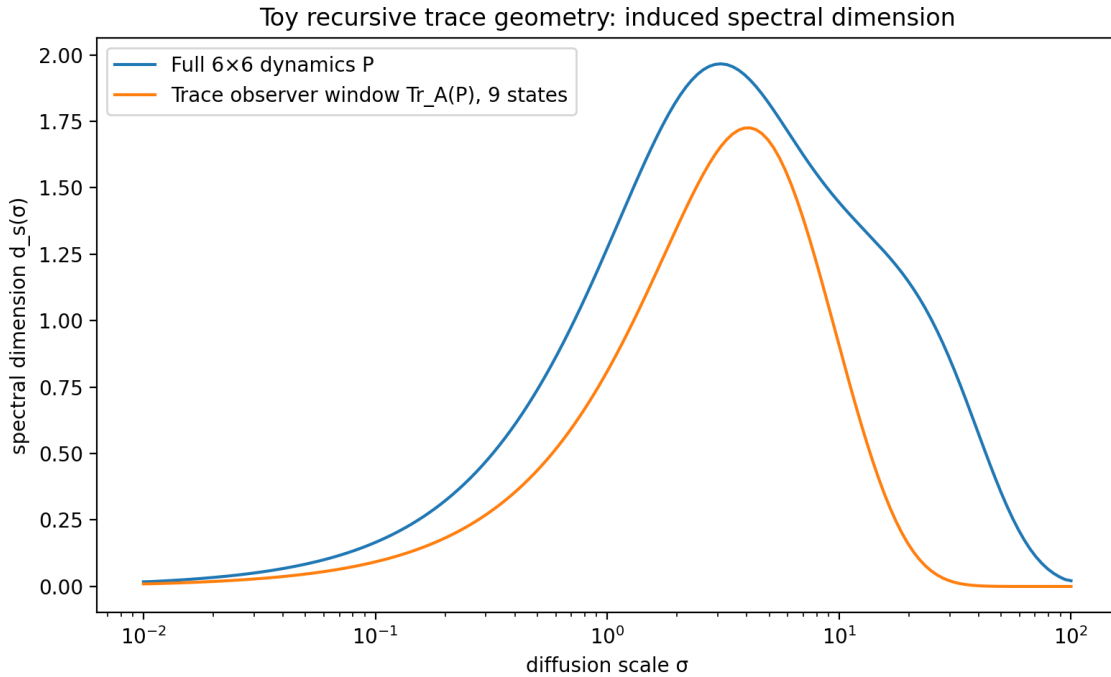


Figure 1: Toy 1: Baseline recursive trace geometry. The observer-window trace induces its own running spectral dimension, distinct from a simple restriction of the full dynamics.

5.2 Toy 2: Hierarchical Recursive Trace Geometry

The second toy model introduced hierarchical modular structure. The full system contained 72 states. A module-level observer window contained 18 states, and a submodule-level observer window contained 6 states.

The resulting spectral curves showed a clear observer-scale dependence: the full hierarchy supported the highest spectral dimension, the module-level trace supported an intermediate spectral dimension, and the submodule-level trace supported a lower spectral dimension.

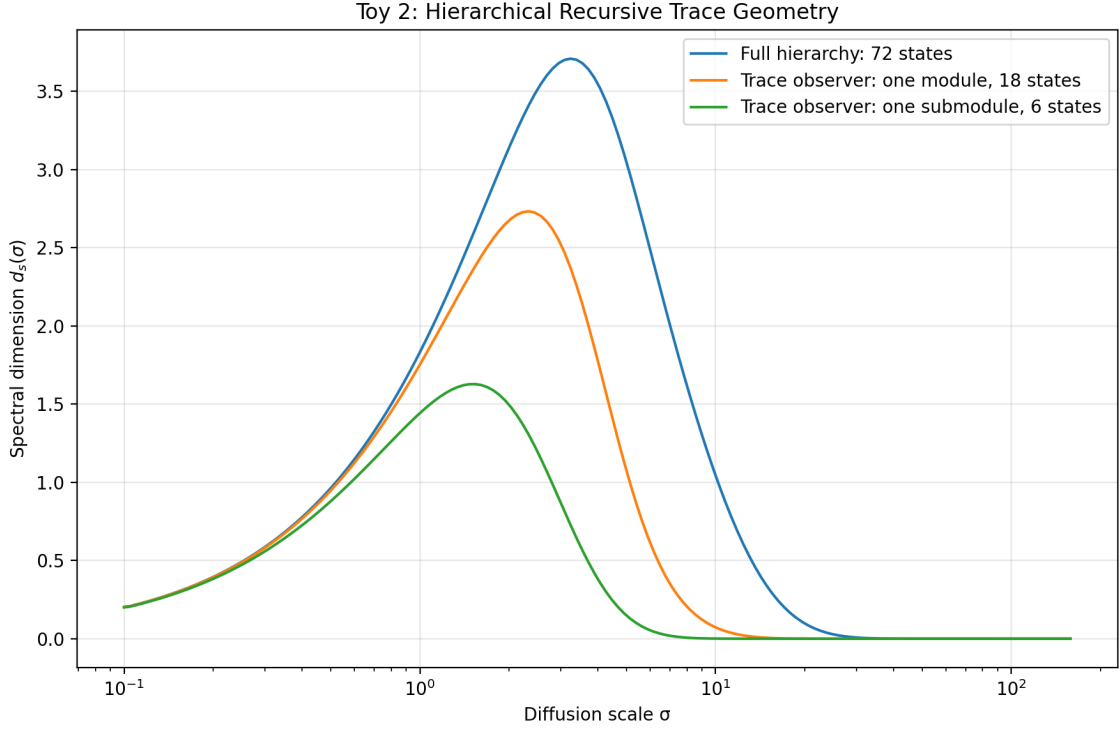


Figure 2: Toy 2: Hierarchical recursive trace geometry. The full hierarchy, module-level observer trace, and submodule-level observer trace exhibit distinct running spectral dimensions. In the representative run, the maximum spectral dimensions were approximately 3.71, 2.73, and 1.63 respectively.

This suggests that observer scale affects the induced spectral geometry.

5.3 Toy 3: Robustness and Controls

The third toy model tested whether the scale-dependent pattern was robust across repeated random realizations and whether hierarchy alone explained the effect. Eighty randomized trials were run for hierarchical systems, modular-control systems, and random-control systems.

Table 1 summarizes the trial averages. Toy 3 was useful precisely because it was humbling: observer scale mattered, but the controls were too similar to justify a strong claim that hierarchy alone caused the effect. This motivated Toy 4, which directly tests trace geometry against ordinary visible subgraph geometry.

5.4 Toy 4: Trace Geometry Versus Visible Subgraph Geometry

Toy 4 is the strongest result. It compares recursive Markov traces directly against ordinary visible-state subgraph restrictions. In ordinary subgraph restriction, only visible-visible transitions are retained. Hidden excursions are discarded. In contrast, recursive trace geometry incorporates

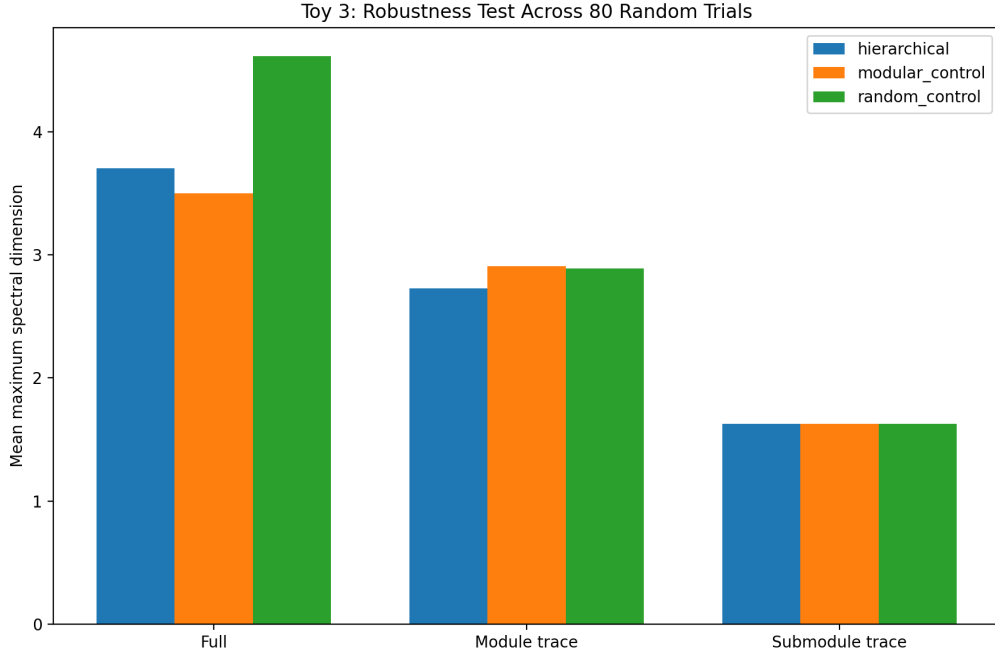


Figure 3: Toy 3: Mean maximum spectral dimensions across 80 randomized trials. The observer-scale pattern is visible, but the controls show that this statistic alone is not sufficient to claim a unique hierarchy effect.

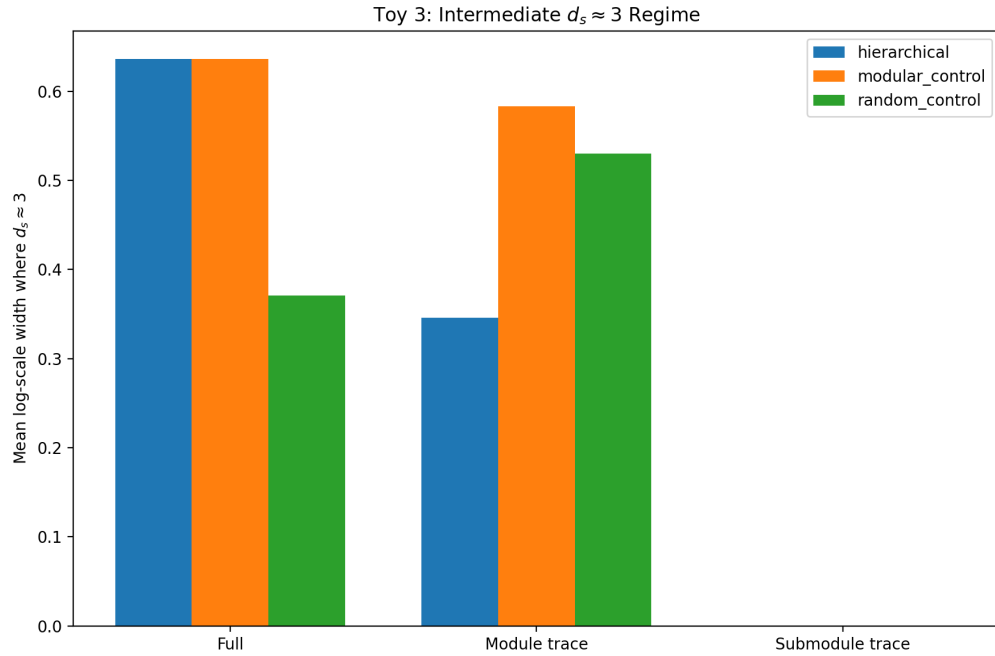


Figure 4: Toy 3: Mean log-scale width for which $d_s \approx 3$. This robustness/control test showed that plateau behavior depends on construction choices and should not be overinterpreted.

Table 1: Toy 3 summary statistics across 80 trials.

Model	Scale	Mean max d_s	Mean width near $d_s = 3$
Hierarchical	Full	3.704	0.636
Hierarchical	Module trace	2.729	0.346
Hierarchical	Submodule trace	1.627	0.000
Modular control	Full	3.502	0.636
Modular control	Module trace	2.907	0.583
Modular control	Submodule trace	1.627	0.000
Random control	Full	4.620	0.371
Random control	Module trace	2.893	0.530
Random control	Submodule trace	1.626	0.000

hidden-state excursions through

$$B(I - C)^{-1}D. \quad (11)$$

Figure 5 shows a representative comparison. At the module-observer scale, the trace observer has a visibly higher running spectral dimension than the ordinary visible-subgraph observer.

Figure 6 summarizes the mean maximum spectral dimensions across repeated randomized hierarchical constructions.

The difference distribution is shown in Figure 7. For the module observer, the trace-minus-subgraph difference is consistently positive.

Table 2: Toy 4 summary statistics across repeated randomized hierarchical constructions.

Scale	Method	Mean max d_s	Std. dev.	Width near $d_s = 3$
Module	Markov trace	2.731	0.008	0.342
Module	Visible subgraph	2.573	0.013	0.000
Submodule	Markov trace	1.627	0.001	0.000
Submodule	Visible subgraph	1.625	0.001	0.000

This demonstrates that recursive trace geometry is not equivalent to simple node deletion or ordinary graph coarse-graining. Hidden dynamics actively reshapes the effective geometry perceived by the observer, especially at the module-observer scale.

6 Mechanism: Hidden Excursions as Effective Geometry

The central mechanism responsible for the observed geometric differences is the hidden excursion term

$$B(I - C)^{-1}D. \quad (12)$$

The factor $(I - C)^{-1}$ is a geometric-series operator encoding excursions of all possible lengths through the hidden sector:

$$(I - C)^{-1} = I + C + C^2 + C^3 + \dots. \quad (13)$$

Consequently, the trace observer dynamics include contributions from arbitrarily long hidden paths before re-entry into the visible observer sector.

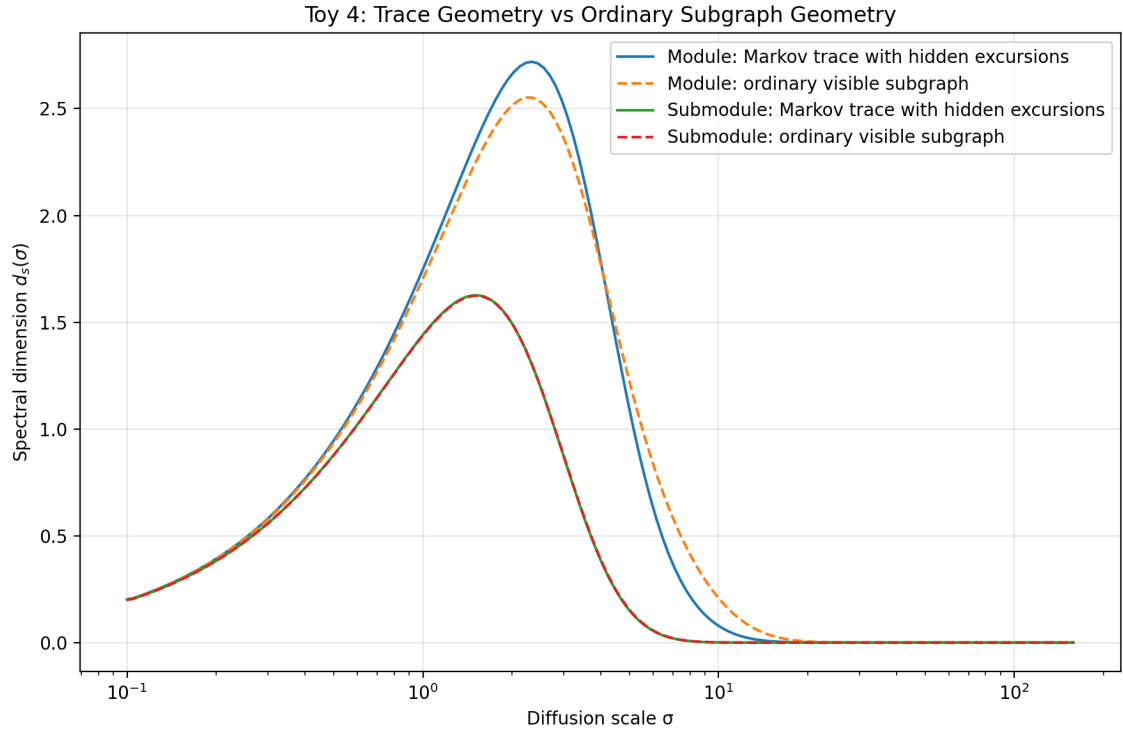


Figure 5: Toy 4: Recursive trace geometry versus ordinary visible-state subgraph geometry. Hidden excursions through inaccessible states modify the running spectral dimension of the visible observer dynamics.

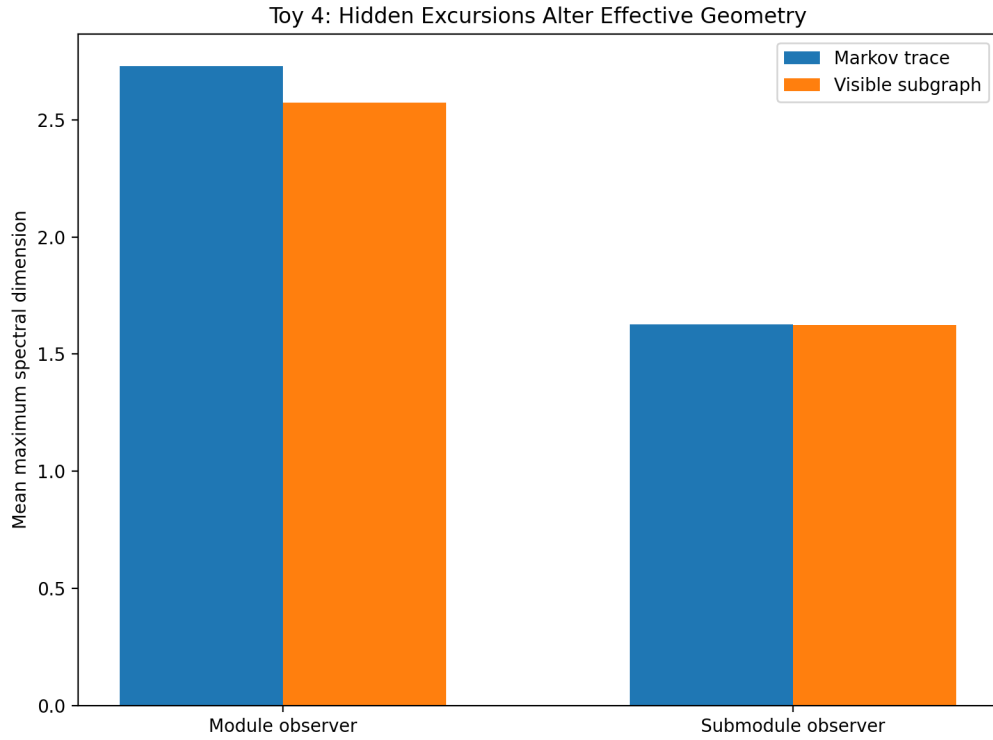


Figure 6: Toy 4: Mean maximum spectral dimension for recursive trace observers and ordinary visible-state subgraph observers. The module-level trace observer consistently exhibits a larger maximum spectral dimension than the corresponding subgraph observer.

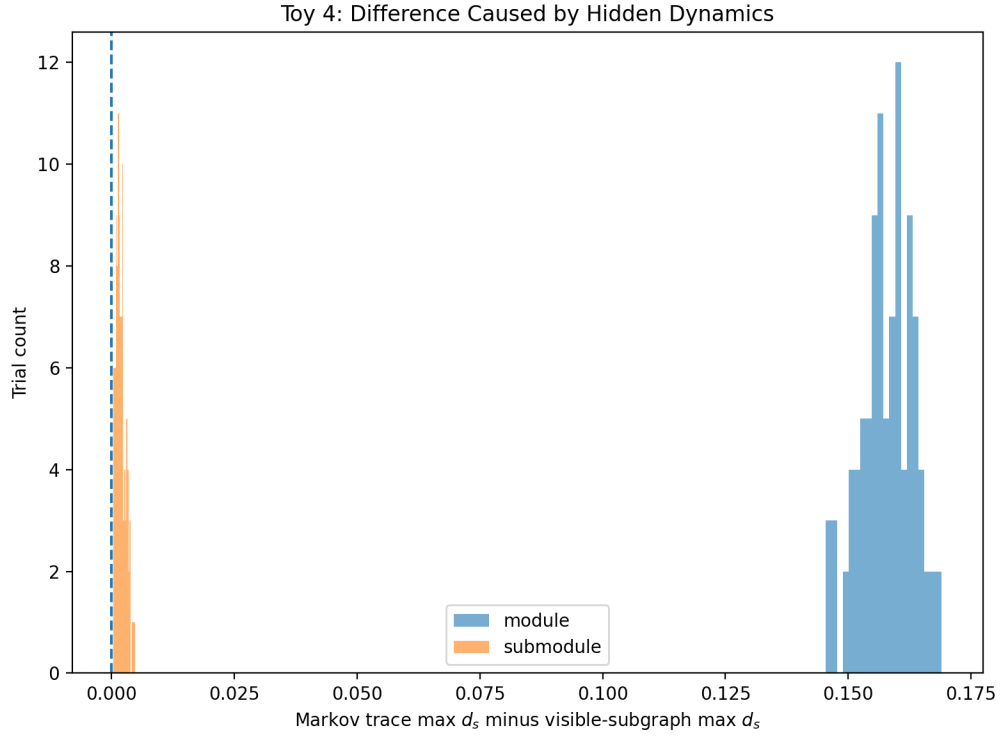


Figure 7: Toy 4: Distribution of trace-minus-subgraph maximum spectral dimension differences across repeated trials. The module-level observer shows a systematic positive shift caused by hidden excursions.

From the perspective of the visible observer, these hidden excursions generate effective long-range couplings between visible states. Ordinary visible-state subgraphs remove these hidden excursions entirely. Recursive trace geometry incorporates the effective connectivity induced by inaccessible hidden dynamics. The hidden sector therefore reshapes observable diffusion structure and alters spectral geometry.

6.1 Minimal Analytical Example

To make the effect explicit, consider a minimal three-state system consisting of two visible states v_1, v_2 together with a hidden state h . Suppose

$$P = \begin{pmatrix} \epsilon & 0 & 1 - \epsilon \\ 0 & \epsilon & 1 - \epsilon \\ \alpha & \alpha & 1 - 2\alpha \end{pmatrix}, \quad (14)$$

where ϵ is a weak direct visible-visible coupling and α controls strong coupling through the hidden sector, with

$$0 < \epsilon \ll 1, \quad 0 < \alpha \leq \frac{1}{2}. \quad (15)$$

The visible observer sector is

$$A = \begin{pmatrix} \epsilon & 0 \\ 0 & \epsilon \end{pmatrix}, \quad (16)$$

while the hidden-sector blocks are

$$B = \begin{pmatrix} 1 - \epsilon \\ 1 - \epsilon \end{pmatrix}, \quad D = \begin{pmatrix} \alpha & \alpha \end{pmatrix}, \quad C = [1 - 2\alpha]. \quad (17)$$

Since C is scalar,

$$(I - C)^{-1} = \frac{1}{2\alpha}. \quad (18)$$

Therefore,

$$B(I - C)^{-1}D = \frac{1 - \epsilon}{2\alpha} \begin{pmatrix} \alpha & \alpha \\ \alpha & \alpha \end{pmatrix} \quad (19)$$

$$= \frac{1 - \epsilon}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}. \quad (20)$$

The resulting trace observer matrix becomes

$$\text{Tr}_A(P) \approx \begin{pmatrix} \epsilon + \frac{1 - \epsilon}{2} & \frac{1 - \epsilon}{2} \\ \frac{1 - \epsilon}{2} & \epsilon + \frac{1 - \epsilon}{2} \end{pmatrix}. \quad (21)$$

For small ϵ , this approaches

$$\begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}, \quad (22)$$

showing that the visible states become strongly coupled through hidden excursions.

The Laplacian of the trace observer is therefore approximately

$$L_{\text{trace}} \approx \begin{pmatrix} 1/2 & -1/2 \\ -1/2 & 1/2 \end{pmatrix}, \quad (23)$$

with eigenvalues

$$\lambda_1 = 0, \quad \lambda_2 = 1. \quad (24)$$

This corresponds to rapid mixing between the visible states induced by hidden excursions.

In contrast, the ordinary visible-state subgraph observer possesses only extremely weak effective coupling between v_1 and v_2 , leading to a near-zero second eigenvalue and correspondingly slow diffusion mixing. Thus the recursive trace observer has a well-defined finite diffusion geometry at intermediate scales, while the visible-state subgraph observer behaves approximately like a disconnected geometry over the same scale regime.

7 Discussion and Limitations

The numerical experiments consistently show that observer scale modifies effective spectral geometry, recursive traces differ systematically from ordinary visible-state subgraphs, hidden excursions increase effective spectral connectivity, and hierarchical observer windows exhibit scale-dependent spectral regimes.

The strongest result is Toy 4: module-level recursive trace observers consistently exhibit larger spectral dimensions than corresponding visible-state subgraph observers across repeated random hierarchical constructions. This supports the interpretation that hidden-state excursions modify observer-relative geometry in a nontrivial way.

The magnitude of this effect depends on the choice of observer window. In the present paper, the primary Toy 4 comparison uses a module-level observer containing 18 visible states. Earlier sparse-observer experiments, using a much smaller cross-module visible window, produced a larger relative separation between trace and subgraph geometries. This suggests that observer sparsity can amplify the hidden-excursion contribution: when the visible window is small, hidden paths supply a greater fraction of the observer’s effective connectivity. The module-level result reported here is therefore the more conservative comparison, while sparse cross-module observers may provide a useful direction for future work.

The present work is exploratory and highly preliminary. It does not derive spacetime, gravity, or quantum theory. It also uses symmetrized diffusion geometry and therefore does not preserve the full directed structure of the original Markov dynamics. Future work should investigate larger systems, directed Laplacians, recursive policies and meta-policies, analytical properties of trace-induced spectra, and possible relationships to information geometry and observer-first approaches in theoretical physics.

8 Conclusion

We have introduced a toy-model framework connecting recursive Markov traces with observer-relative spectral geometry. Using a sequence of four toy models, we showed that hidden-state excursions can modify induced spectral geometries relative to ordinary visible-state subgraphs.

The central proposal is:

Recursive trace logic supplies the observer hierarchy; spectral geometry supplies the emergent spatial structure of each observer’s world.

While preliminary, the framework provides a mathematically tractable bridge between recursive observer systems and emergent relational geometry.

Acknowledgments

The recursive trace logic framework discussed in this paper is motivated partly by publicly available seminar discussions by Donald Hoffman and collaborators. Since this framework remains unpublished work in progress, it is treated here as conceptual motivation rather than as a formal technical source.

A Reproducibility Notes

The toy models were implemented using finite row-stochastic Markov matrices. For a visible observer set A , the Markov trace was computed by partitioning P into visible and hidden blocks and evaluating

$$\mathrm{Tr}_A(P) = A + B(I - C)^{-1}D. \quad (25)$$

The ordinary subgraph control was computed by retaining visible-visible transitions and renormalizing rows. For spectral geometry, transition matrices were symmetrized, converted to normalized Laplacians, and evaluated using eigenvalue-based heat return probabilities.

All figures and summary CSV files are included in the accompanying package.

References

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